📌 Royal Mail Group Project – Interview Explanation

🔹 Project Overview

I worked on a **data analytics project** for Royal Mail Group where we integrated data from multiple systems like **Oracle Siebel, Oracle Service Cloud, and Parcelforce** into **MySQL** and **Excel**, and then created dashboards in **Power BI** to track delivery performance, customer service, and financial KPIs.

🔹 Data Sources

- **Oracle Siebel CRM** → Customer details, service requests, complaints.
- **Oracle Service Cloud** → Ticket volumes, response times, resolution rates, SLA compliance, customer feedback.
- **Parcelforce** → Parcel tracking, delivery times, delivery success/failure, route optimization data.
- **Excel Files** → Financial data like revenue, costs, operational expenses, and customer survey results.

🔹 ETL Process

1. **Extract**
   - Pulled raw data from Siebel, Service Cloud, and Parcelforce (via APIs and flat files).
   - Imported into staging tables in **MySQL**.
   - Uploaded Excel files manually into Power BI for financial data.
2. **Transform (In MySQL)**
   In MySQL, we performed data cleaning and transformation:
   - **Data Cleaning** → Removed duplicates, handled NULL values, corrected date/time formats.
   - **Standardization** → Unified status values like "Delivered", "delivered", "D" into one value.
   - **Business Rules** → Marked SLA_Breached = Yes if resolution time > 48 hrs.
   - **Joins & Aggregations** → Merged data across systems for consolidated fact tables.
   - **Pre-aggregation** → Created daily/weekly summary tables for faster reporting.
3. **Load**
   - Loaded final cleaned data into **fact and dimension tables** in MySQL.
   - These were directly connected to **Power BI** for reporting.

- Excel data was linked in Power BI for additional financial KPIs.

🔹 **SQL Queries Written**

✅ **Example – Region-wise delivery performance**

sql

```
SELECT r.RegionName,
    COUNT(d.ParcelID) AS Total_Parcels,
    SUM(CASE WHEN d.Status = 'Delivered' THEN 1 ELSE 0 END) AS Delivered,
    SUM(CASE WHEN d.Status = 'Delayed' THEN 1 ELSE 0 END) AS Delayed
FROM Fact_Delivery d
JOIN Dim_Region r ON d.RegionID = r.RegionID
GROUP BY r.RegionName;
```

✅ **Example – Average ticket resolution time**

sql

```
SELECT AVG(TIMESTAMPDIFF(HOUR, CreatedDate, ResolvedDate)) AS Avg_Resolution_Hours
FROM Fact_Tickets;
```

✅ **Example – SLA breach count**

sql

```
SELECT COUNT(*) AS SLA_Breaches
FROM Fact_Tickets
WHERE SLA_Breached = 'Yes';
```

🔹 **Data Model (Tables & Columns)**

Fact Tables

- **Fact_Delivery**
  - ParcelID (INT)
  - DeliveryDate (DATE)
  - Status (VARCHAR)
  - RegionID (INT, FK)
  - CustomerID (INT, FK)
- **Fact_Tickets**

    - TicketID (INT)
    - CreatedDate (DATETIME)
    - ResolvedDate (DATETIME)
    - AgentID (INT, FK)
    - SLA_Breached (VARCHAR)
- **Fact_Financials**
    - ReportDate (DATE)
    - Revenue (DECIMAL)
    - Cost (DECIMAL)
    - Profit (DECIMAL)
    - RegionID (INT, FK)

Dimension Tables

- **Dim_Date** → DateKey, Date, Month, Quarter, Year
- **Dim_Customer** → CustomerID, Name, Segment, RegionID
- **Dim_Region** → RegionID, RegionName, Country
- **Dim_Agent** → AgentID, Name, Department

🔹 **KPIs Designed**

Delivery KPIs

- On-Time Delivery % = Delivered on time / Total Deliveries
- Average Delivery Time (days/hours)
- Delayed Deliveries %

Customer Service KPIs

- Average Resolution Time (hours)
- First Response Time (hours)
- SLA Compliance %
- Ticket Volume (per agent, per region)

Financial KPIs

- Revenue by Region
- Cost per Parcel
- Profit Margin %

- Year-over-Year Growth

- **Power BI Development → Deployment**

1. **Power BI Desktop**
    - Connected to MySQL & Excel.
    - Built star-schema model with relationships.
    - Created DAX measures for KPIs.
    - Designed interactive dashboards for Delivery, Customer Service, and Finance.
2. **Power BI Service – Dev to Prod**
    - Reports first published to **Dev Workspace** for testing.
    - Used **Deployment Pipelines** to move from **Dev → Test → Prod**.
    - In Prod, only business-approved datasets & dashboards were published.
3. **Publishing to Client**
    - Reports were packaged into a **Power BI App**.
    - Shared with stakeholders with **role-based access** (view-only for business users, edit for analysts).
    - Clients accessed dashboards via **Power BI Service web portal or mobile app**.

- **Business Impact**

- Created a **single version of truth** for delivery, customer support, and financial data.
- Helped **reduce delivery failures** by identifying regional bottlenecks.
- Improved **SLA compliance** by tracking agent performance.
- Automated financial reporting → saving hours of manual work every month.

My Role and Responsibilities and Client Requirements: -

- **Client Requirements (What they asked for)**

The client (Royal Mail stakeholders) wanted:

1. **Single source of truth** → Combine data from Oracle Siebel, Service Cloud, Parcelforce, and Finance Excel files into one reporting system.
2. **Operational dashboards** → Track parcel delivery times, success rate, and bottlenecks by region.
3. **Customer service dashboards** → Monitor ticket volumes, response times, SLA compliance, and customer feedback.
4. **Financial dashboards** → Show revenue, cost, and profit trends (monthly, quarterly, yearly).
5. **User-friendly Power BI reports** → With filters (region, time period, delivery type, customer segment).
6. **Automated refresh** → So reports always have latest data without manual intervention.
7. **Secure access** → Role-based sharing (e.g., executives see all data, regional managers only see their area).

🔹 **My Roles & Responsibilities**

I worked as a **Data Analyst / Power BI Developer** and was responsible for:

- **Requirement gathering** → Interacted with business users to understand what KPIs they wanted.
- **ETL Process**
  - Worked with MySQL team to clean, transform, and prepare reporting tables.
  - Wrote SQL queries for aggregations, joins, and business rules.
- **Data Modeling in Power BI**
  - Created star schema (fact & dimension tables).
  - Built relationships and optimized data model.
- **Report Development**
  - Created DAX measures (e.g., SLA compliance %, Avg Resolution Time, Delivery Success Rate).
  - Designed dashboards with slicers, drillthrough, and bookmarks.
- **Testing & Validation**
  - Validated numbers against MySQL queries & Excel files before publishing.
- **Deployment**
  - Published reports to Power BI Service (Dev → Test → Prod pipeline).
  - Configured scheduled refresh.

- **Client Support**
  - Created Power BI App and shared with clients.
  - Handled change requests (new KPI, new filter, layout changes).

🔹 **My Daily Tasks (Real-Time Work)**

Here's what my **day-to-day work** looked like:

1. **Morning Check**
   - Verified scheduled Power BI refresh (data pipelines working correctly).
   - Checked for any failures in MySQL jobs or Excel uploads.
2. **Data Preparation (SQL/MySQL)**
   - Wrote queries to:
     - Remove duplicates and NULLs.
     - Create business rules (e.g., SLA_Breached flag, Delivery_Success calculation).
     - Aggregate daily/weekly KPIs.
   - Example: calculating average delivery time or SLA compliance.
3. **Power BI Development**
   - Connected Power BI to MySQL and Excel.
   - Created new **measures in DAX** (like YoY growth, % of delayed deliveries).
   - Updated visuals or added new filters based on client feedback.
4. **Client Meetings / Stand-ups**
   - Attended daily stand-up with the team → discussed yesterday's progress, today's plan, and blockers.
   - Weekly meeting with business users to demo dashboards and collect feedback.
5. **Testing & Validation**
   - Compared Power BI dashboard results with raw SQL query output for accuracy.
   - Ensured KPIs matched client definitions.
6. **Deployment**
   - After approval, moved report from **Dev → Test → Prod**.
   - Packaged in **Power BI App** and shared with users.
7. **Ad-hoc Tasks**
   - Handled client requests (e.g., "Can we add a filter by parcel type?").

- Troubleshot issues (like refresh failures, missing data, incorrect numbers).

🔹 **How I Can Answer in Interview (Sample Script)**

👉 "In my project at Royal Mail Group, the client wanted a single reporting system combining customer service, delivery, and financial data. I worked as a Data Analyst / Power BI Developer. My responsibilities included extracting data from Oracle Siebel, Service Cloud, and Parcelforce into MySQL, performing data cleaning and transformations using SQL, and then building interactive dashboards in Power BI. I created KPIs like SLA compliance %, delivery success rate, and revenue trends.

On a daily basis, I checked data refresh jobs, wrote SQL queries to clean and aggregate data, built DAX measures, and developed dashboards. I also attended stand-ups, validated numbers against source systems, and handled client change requests. Once validated, I deployed reports from Dev → Test → Prod using Power BI Service and shared them with stakeholders via Power BI Apps. This gave management real-time insights into customer satisfaction, operational efficiency, and financial performance."

📊 Charts & Visuals Used in our project

1. KPI Cards / Multi-row Cards

- **Use case**: Show at-a-glance metrics.
- **What I showed**:
  - Total Tickets, Open Tickets, Closed Tickets.
  - SLA Compliance % (Target vs Actual).
  - Delivery Success Rate %.
  - Monthly Revenue, Cost, Profit %.

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2. Clustered Column Chart / Bar Chart

- **Use case**: Compare values across categories.
- **What I showed**:
  - Tickets by Category (Technical, Billing, Delivery Issues).
  - Parcels delivered vs delayed by Region.
  - Revenue by Product Type or Service Line.

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3. Line Chart

- **Use case**: Show **trends over time**.

- **What I showed**:
    - Ticket Volume trend (daily/weekly/monthly).
    - Average Response Time trend.
    - Revenue & Cost trends over months.
    - Delivery Success Rate trend.

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4. Stacked Column / 100% Stacked Bar

- **Use case**: Show distribution or contribution.
- **What I showed**:
    - SLA Met vs SLA Breached tickets by Department.
    - Parcel delivery status (On-time, Delayed, Lost).
    - Customer feedback (Positive, Neutral, Negative).

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5. Donut / Pie Chart (used sparingly – only for % share)

- **Use case**: Show proportion of a whole.
- **What I showed**:
    - % of tickets by Priority (High, Medium, Low).
    - % of parcels by Service Type (Standard, Express, International).

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6. Map / Filled Map

- **Use case**: Show regional performance.
- **What I showed**:
    - Delivery Success Rate by Region / City.
    - Customer Complaints by Location.
    - Revenue contribution by Region.

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7. Table / Matrix

- **Use case**: Detailed view with drill-down.
- **What I showed**:
    - Ticket details with Agent, Status, Resolution Time.
    - Region → City → Postcode delivery performance.

    - Financial breakdown by Department & Month.

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8. Gauge / Bullet Chart

- **Use case**: Show progress vs Target.
- **What I showed**:
    - SLA Compliance % vs Target 95%.
    - Delivery Success % vs Target 98%.
    - Cost Efficiency % vs Target.

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9. Decomposition Tree (Power BI AI Visual)

- **Use case**: Drill into reasons for performance.
- **What I showed**:
    - Why deliveries got delayed → by Region → by Route → by Weather / Traffic / Operational Issue.
    - Why tickets breached SLA → by Agent → by Category → by Priority.

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10. Slicers / Filters

- **Use case**: Interactivity.
- **What I gave users**:
    - Date Range (Month, Quarter, Year).
    - Region / City.
    - Ticket Category / Priority.
    - Service Type (Standard, Express, International).

🎯 **Example Answer for Interview**

👉 *"In my project, I used a mix of visuals depending on the KPI. For trends like ticket volumes or delivery success rate, I used line charts. For comparing regions or categories, I used bar/column charts. KPI cards were used for at-a-glance metrics like SLA compliance %, delivery success %, and revenue. For geographical analysis, I used filled maps to show performance by region. I also used decomposition trees for root-cause analysis of delays. Finally, I created drill-through tables for detailed record-level analysis and slicers for interactivity, so users could filter by time, region, or service type."*